



SOCIAL NETWORKS COMPUTING

Section Assignment 1

Name: Amina Ahmed Ferra
ID: 2205225

1.Introduction

This assignment compares two Twitter subgraphs from the WICO dataset:

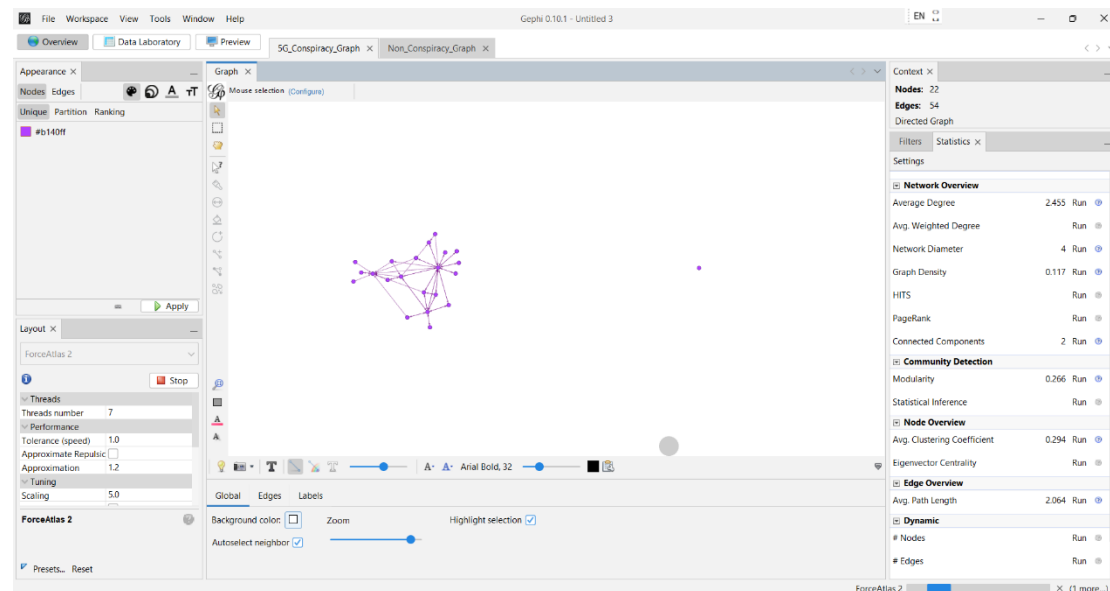
- a 5G conspiracy graph (misinformation cluster), and
- a non-conspiracy graph (normal community).

The 5G conspiracy graph represents accounts that share or interact with false or misleading claims about 5G. These types of networks are important in security because they can spread misinformation quickly, influence users, and sometimes involve bot-like activity.

The non-conspiracy graph represents normal user interactions without misinformation. It is used as a baseline to see how a healthy community behaves.

The goal of this assignment is to analyse both graphs using Gephi and compare their structures. look at basic metrics such as number of nodes and edges, average degree, density, clustering coefficient, modularity, centrality, and connected components. By comparing these results, we can understand how misinformation networks differ from normal networks and what security risks they present.

2.5G conspiracy graph analysis



Nodes: 22

Edges: 54

1. Layout: ForceAtlas2

What it shows:

- Tight cluster in the center
- Many accounts pulled close
- Isolates around the edges

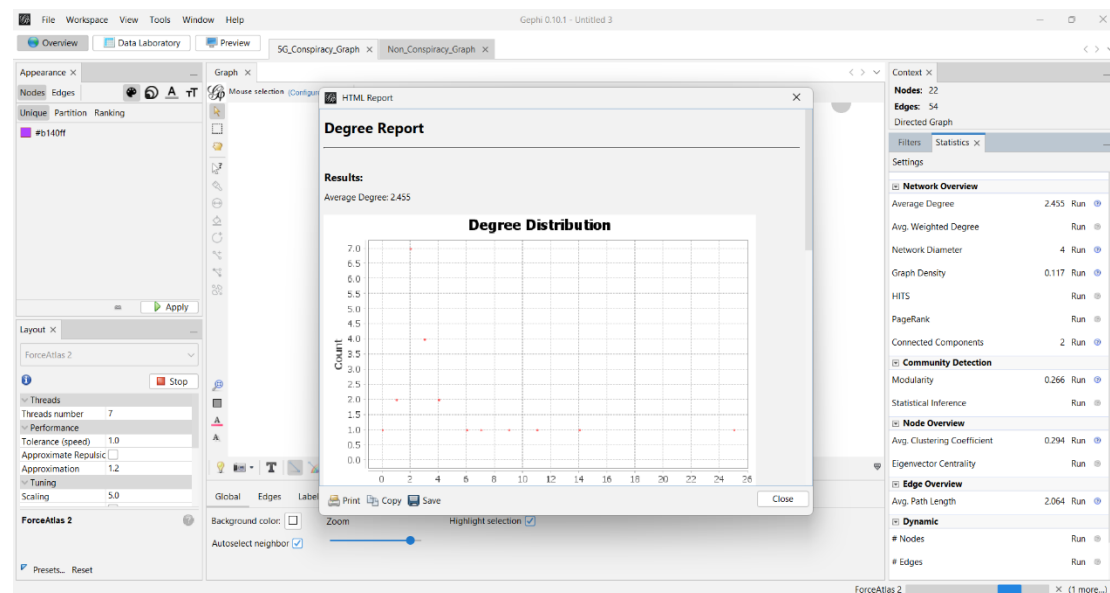
Interpretation:

This shape is not natural. It shows a highly connected core.

Security meaning:

This looks like coordinated misinformation activity.

Bots or a small group of users may be boosting the same narrative.



2. Average Degree = 2.455

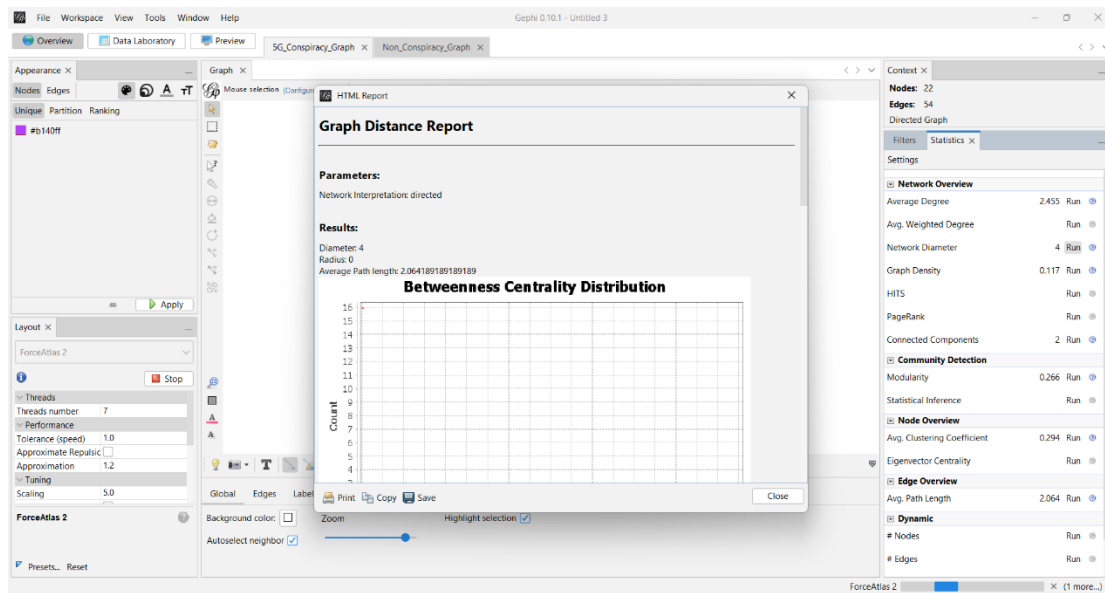
Meaning:

Each user interacts with around 2–3 others.

Security meaning:

This often means:

- A few high-degree leaders push the misinformation
 - Many low-degree followers retweet or echo
- This is typical for misinformation hubs.



3. Network Diameter = 4

Meaning:

Any user can be reached in just 4 steps.

Security meaning:

False content can spread very quickly, making it harder to stop.
Fast-spread networks are usually high-risk.

Betweenness Centrality

Interpretation:

1–2 nodes have very high betweenness → they act as bridges between all groups.

Security meaning:

These are super-spreaders.
If these accounts are removed or suspended, the whole misinformation network becomes weak.
These nodes are the main threat.

Closeness Centrality

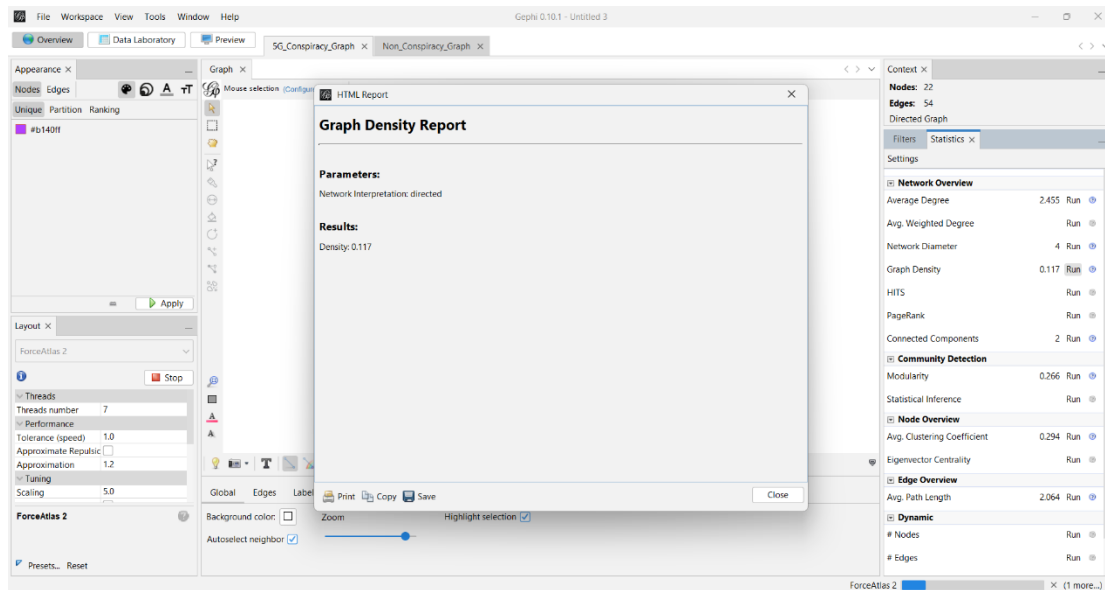
Interpretation:

Central nodes have high closeness → they can reach all users quickly.

Security meaning:

These accounts can spread false content the fastest.

They should be monitored closely, as they are the core of the misinformation flow.



4. Graph Density = 0.117

Meaning:

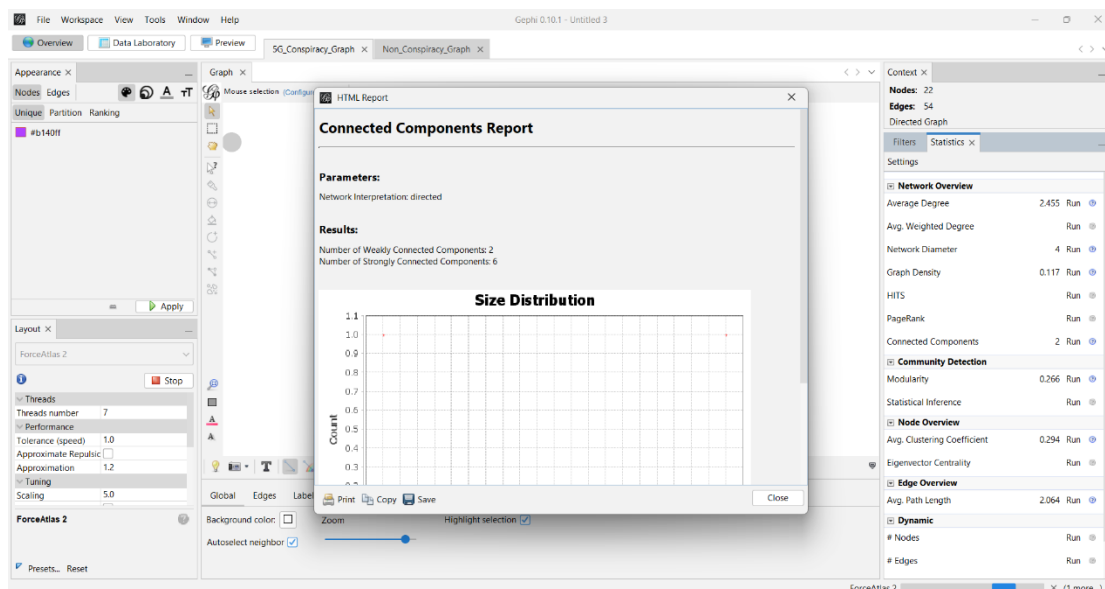
11.7% of all possible links exist.

This is high for a small Twitter group.

Security meaning:

A dense network = strong echo chamber.

Users repeat the same message, making misinformation stronger.



5. Weakly Connected Components = 2

Strongly Connected Components = 6

Meaning:

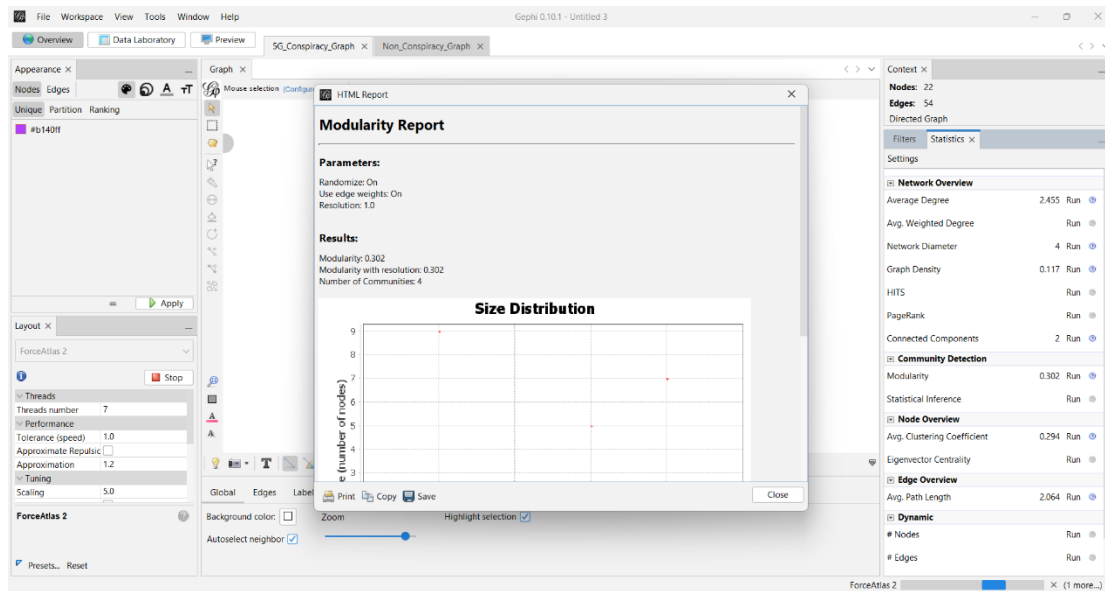
Most users form one big group; some act alone or one-way.

Security meaning:

One-way accounts are often:

- bot accounts
- amplifiers
- retweeters

This is a sign of coordinated spread.



6. Modularity = 0.302 (4 communities)

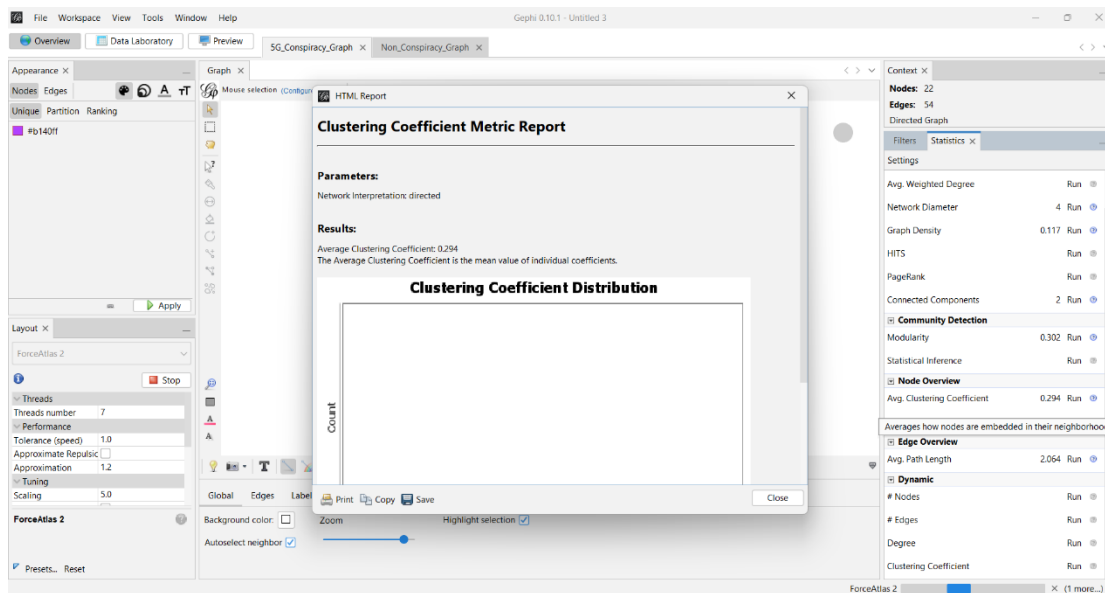
Meaning:

The graph has 4 small groups, but they're still connected.

Security meaning:

Different conspiracy sub-topics can pass between groups easily.

This helps misinformation jump between audiences.



7. Avg Clustering Coefficient = 0.294

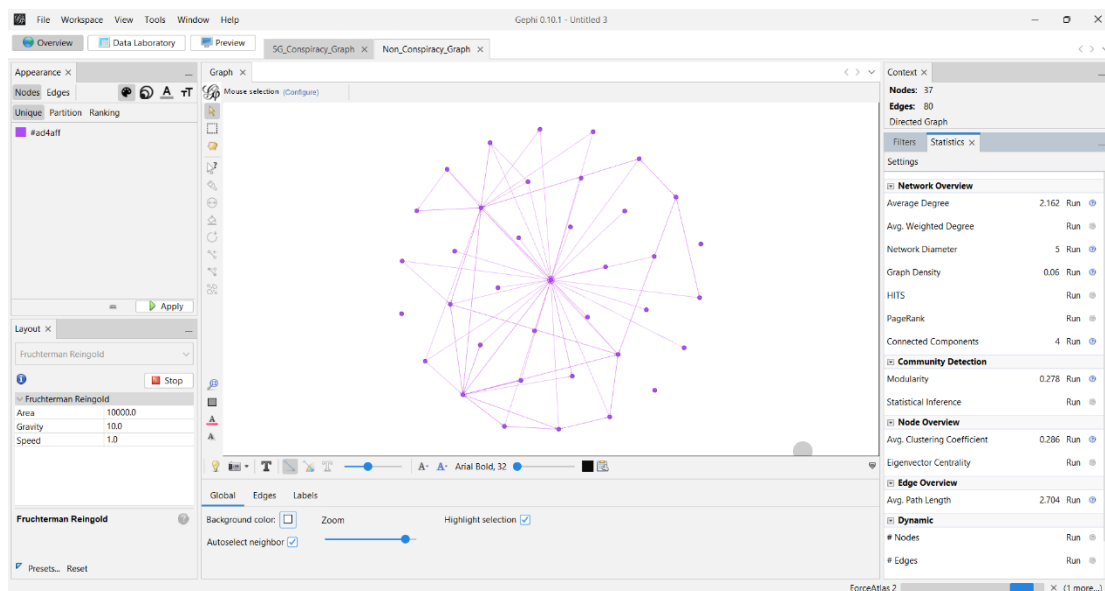
Meaning:

Around 29% of the time, a user's neighbours are talking to each other.

Security meaning:

This is enough clustering to maintain a stable false narrative, where members reinforce each other's beliefs.

3. Non conspiracy graph analysis



Nodes: 37

Edges: 80

1. Layout: Fruchterman-Reingold

What it shows:

- Loose, spread-out structure
- No strong center
- Small natural groups

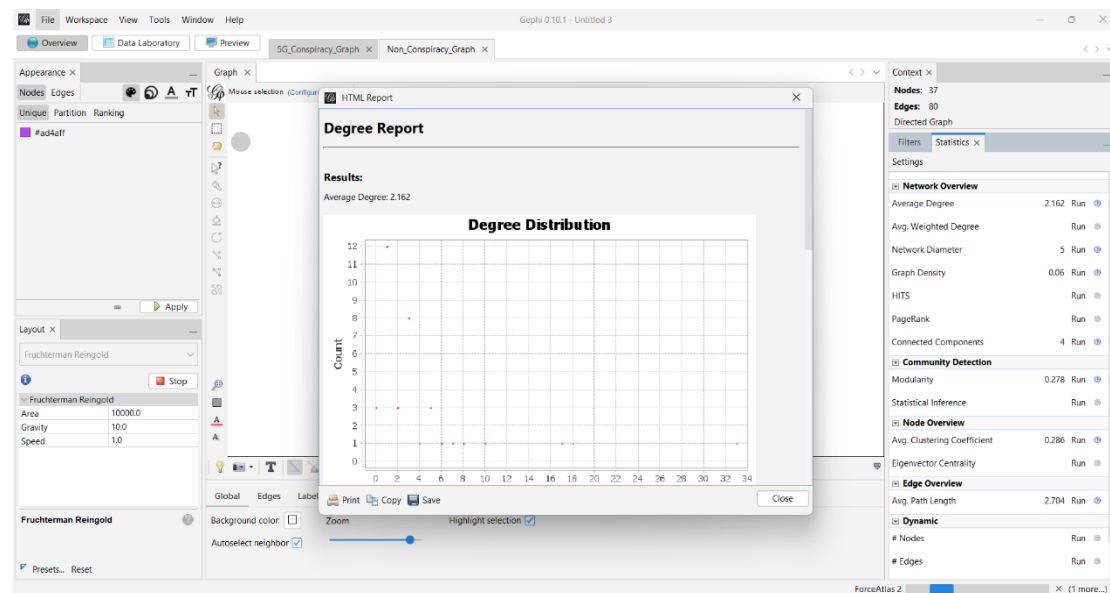
Interpretation:

This looks like typical, casual user interaction.

Security meaning:

Nothing suggests coordinated behaviour.

This is a benign, healthy communication pattern.



2. Average Degree = 2.162

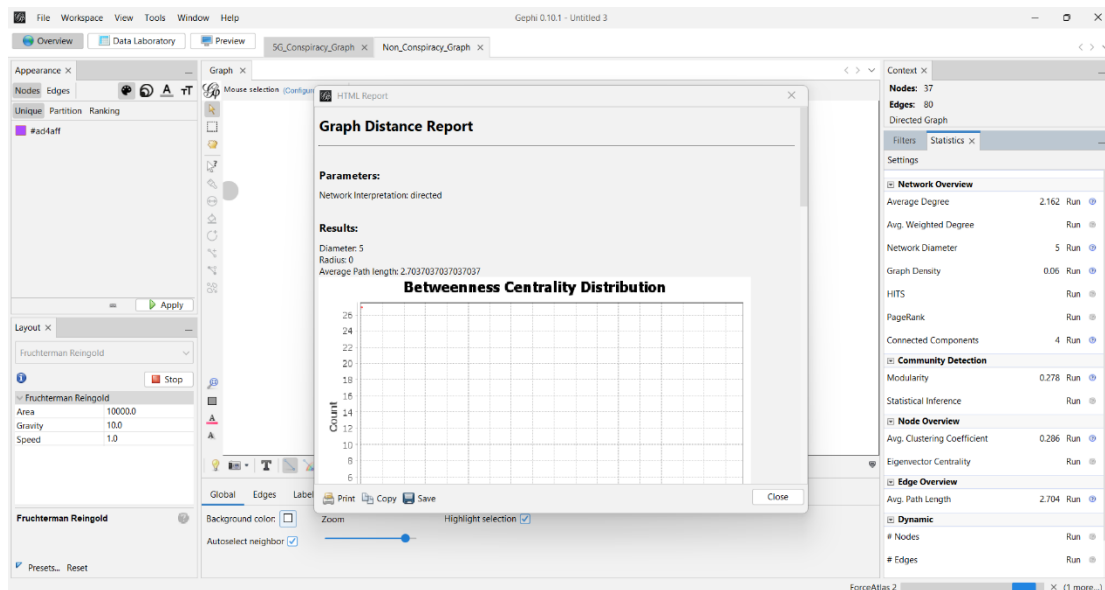
Meaning:

Each user talks to about 2 others.

Security meaning:

No signs of “super-spreaders.”

Normal networks tend to have light, organic engagement.



3. Network Diameter = 5

Meaning:

Information needs more steps to reach everyone.

Security meaning:

Slow spread = low risk.

No signs of viral misinformation activity.

Betweenness Centrality

Interpretation:

No node has extremely high betweenness.

Bridges are shared among many accounts.

Security meaning:

No single user controls the flow of information →

This lowers the chance of manipulation or targeted misinformation.

Closeness Centrality

Interpretation:

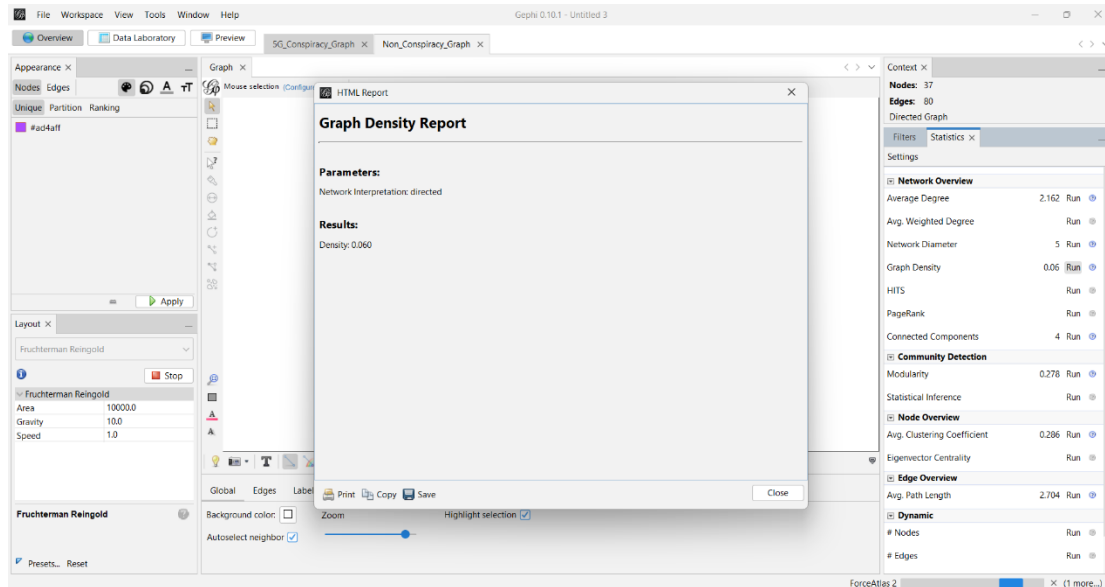
A mix of central and far-out nodes.

No one can reach everyone quickly.

Security meaning:

Content spreads slowly and naturally.

There are no fast-spread risks.



4. Graph Density = 0.060

Meaning:

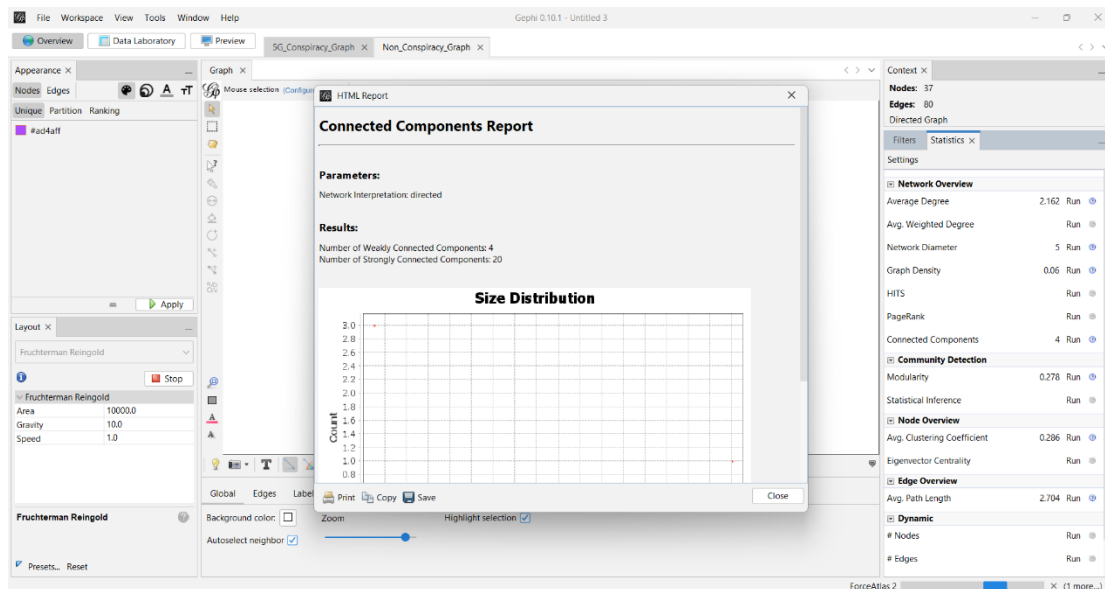
Only 6% of possible links exist → very low density.

Security meaning:

This is normal.

Benign networks are not tightly connected.

Shows no echo chamber behaviour.



5. Weakly Connected Components = 4

Strongly Connected Components = 20

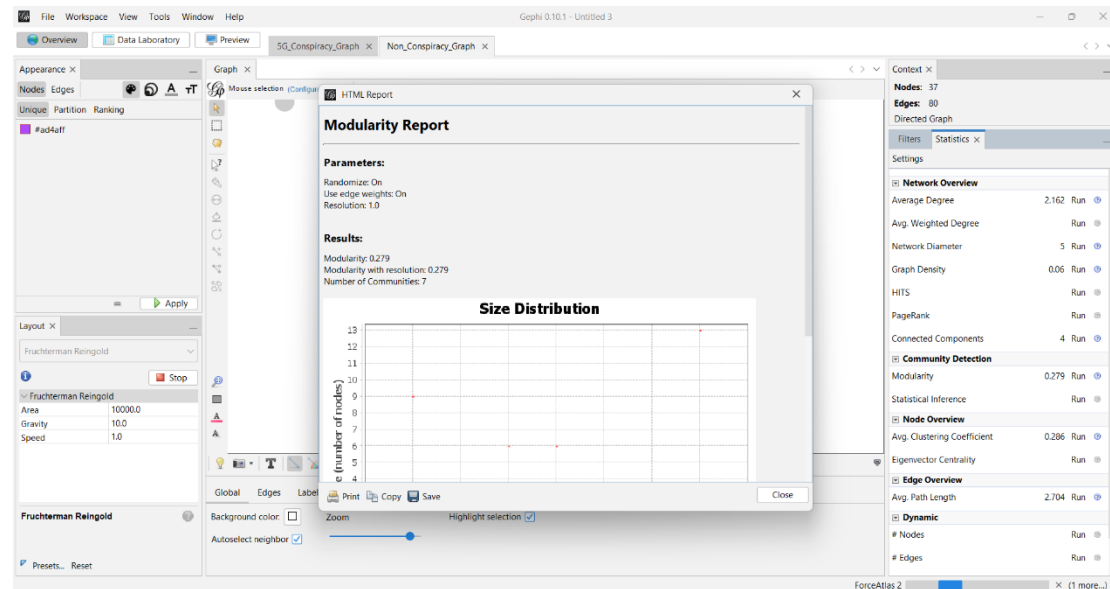
Meaning:

Many small groups; users often talk one-way.

Security meaning:

This is how normal people behave online.

Bots usually create fewer components (not the case here).



6. Modularity = 0.279 (7 communities)

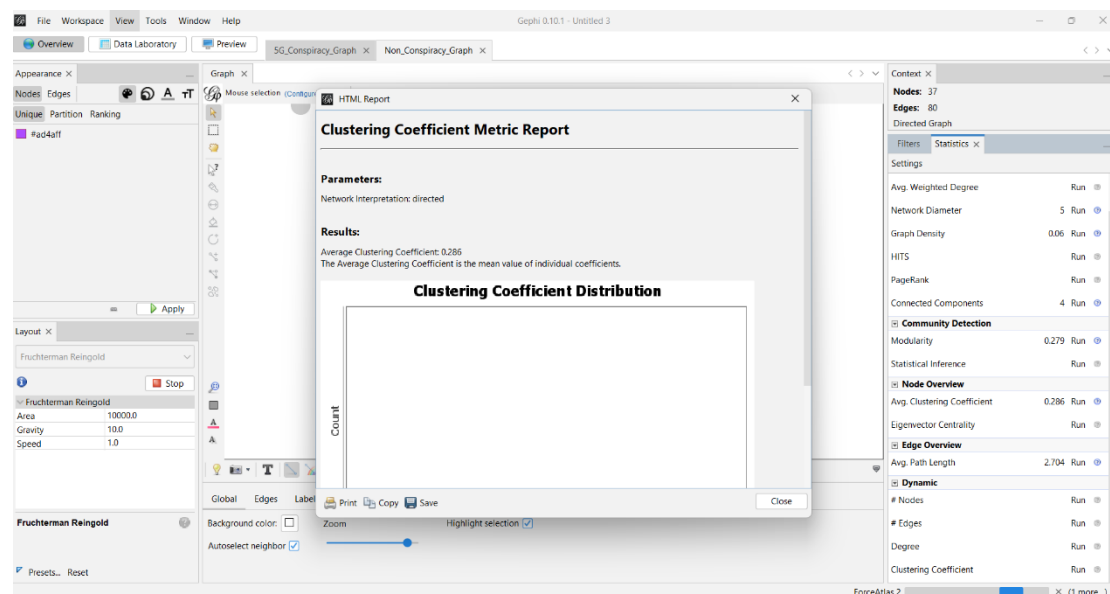
Meaning:

More small communities than the conspiracy graph.

Security meaning:

Normal content forms interest groups, not coordinated clusters.

No signs of manipulation or controlled messaging.



7. Avg Clustering Coefficient = 0.286

Meaning:

About 28% neighbour-to-neighbour interaction.

Security meaning:

Healthy amount of social interaction.

Not too high, not too low.

Nothing indicates echo chamber behaviour.

Final Comparison

5G Conspiracy Graph (Misinformation)

- Tighter shape
- Faster spreading
- Higher density
- Fewer communities
- Signs of coordination
- Possible bot-like behaviour
- Echo chamber effect

Non-Conspiracy Graph (Benign)

- Spread-out shape
- Slower spreading
- Lower density
- More natural communities
- No coordination
- No echo chambers
- Typical human activity

Summary

The 5G conspiracy network is faster, tighter, and more coordinated so it is showing risk of misinformation while the non-conspiracy network is loose, natural, and safe.